

UNIT 3: COMMUNICATION SKILLS FOR PRESENTATION – WRITING, DESIGNING & SPEAKING

Unit 3: Communication Skills for Presentation: Writing, Designing, and Speaking Thesis and Project Report Writing, Technical Proposal Writing, How to Pitch an Idea: Process, Preparation and Structure, Elements of Speech Delivery: Passion, Poise & Illustrations.

1. Thesis and Project Report Writing

♦ What is a Thesis?

A **thesis** is a long, formal piece of writing based on **original research or investigation**. It is usually written by students of higher studies (MTech, PhD) to **prove their understanding and contribution** to a subject.

Example:

A student writes a thesis titled “**Prediction of Heart Disease using Machine Learning Algorithms.**”

It includes data collection, analysis, model design, and final results.

♦ What is a Project Report?

A **project report** describes the **planning, process, implementation, and results** of a project.

It is more practical and is commonly required in **B.Tech / engineering projects**.

Example:

A B.Tech student writes a **project report** on “Smart Irrigation System using IoT.”

It includes circuit design, code, sensor data, and working output.

♦ Purpose of Thesis / Project Report

- To communicate your **research or project work clearly**.
- To present **data, methods, and conclusions** systematically.
- To show your **understanding of engineering applications**.
- To provide a **record** for academic or professional use.

◇ Structure of a Good Thesis / Project Report

Section	Content Description	Example (Topic: Face Recognition System)
1. Title Page	Title, name, roll number, department, college, guide name.	“Face Recognition Attendance System using Python”
2. Certificate	Signed approval from guide or department.	Signed by project supervisor.
3. Acknowledgment	Thanks to mentors, friends, or family.	“We thank our guide Prof. Sharma...”
4. Abstract	150–250 words summary of aim, method, and results.	“This project automates attendance using OpenCV face detection.”
5. Table of Contents	List of chapters and page numbers.	—
6. Introduction	Background, objectives, and importance of the topic.	“Attendance management is time-consuming in schools...”
7. Literature Review	What others have already done.	“Previous studies used manual detection...”
8. Methodology	Steps followed, tools used, and approach.	“Used Python, OpenCV, and SQLite database.”
9. Implementation / Work Done	Coding, experiments, testing, screenshots.	—
10. Results & Discussion	Tables, charts, and observations.	“Accuracy achieved: 93%.”
11. Conclusion & Future Scope	Summary and improvements possible.	“Can be integrated with RFID for better security.”
12. References	Books, websites, or papers cited.	IEEE format preferred.
13. Appendix	Extra code or datasets.	—

◇ Writing Tips

- ✓ Use short, clear sentences.
- ✓ Maintain **past tense** and **third person** tone (“It was observed that...”).
- ✓ Include **figures, tables, and charts** for better clarity.
- ✓ Number sections properly (1.0, 1.1, etc.).

◇ Common Mistakes

- ✗ Copy-paste from internet (plagiarism).
- ✗ Poor formatting and grammar.
- ✗ Missing objectives or results.
- ✗ No references or credits.

2. Technical Proposal Writing

◇ What is a Technical Proposal?

A **technical proposal** is a **written plan** that explains an idea or project **before it is built**.

It is used to **convince others** (teachers, investors, or companies) to approve or fund your project.

Example:

You write a proposal titled **“Smart Helmet with Accident Detection and GPS Alert.”**

You explain the problem (bike accidents), your solution (smart helmet), cost, and expected results.

◇ Purpose

- To **present a plan** for solving a problem.
- To **gain support** or approval for implementation.
- To **demonstrate feasibility** and technical knowledge.

◇ Types of Technical Proposals

Type	Description	Example
Solicited Proposal	Requested by an organization or teacher.	“Submit a proposal for a college tech fest project.”
Unsolicited Proposal	Sent voluntarily by the proposer.	Startup student sends idea to an investor.

◇ Structure of a Technical Proposal

Section	Description	Example (AI Waste Sorting System)
1. Title Page	Title, proposer details, institution.	“AI-Powered Waste Sorting System”
2. Abstract	Short summary of idea.	“An ML model to separate dry/wet waste automatically.”

3. Introduction	Explain background & problem.	“Manual waste sorting is slow and unhygienic.”
4. Objectives	Main goals of project.	“To automate waste segregation.”
5. Proposed Solution / Methodology	Approach, technology used.	“Uses camera + ML model for classification.”
6. Expected Outcome	What results or benefits you expect.	“Reduce manual effort and improve recycling.”
7. Time & Cost Estimation	Timeline and budget plan.	“Duration: 3 months; Cost: ₹10,000.”
8. Conclusion	Final convincing summary.	“This system promotes clean cities using AI.”
9. References	Books, papers, or websites used.	IEEE format.

♦ Qualities of a Good Proposal

- Clear and **logical structure**.
- **Facts and data-based**.
- **Formal and concise** writing.
- **Realistic timeline and budget**.

♦ Common Mistakes

- ✗ Overly technical without clarity.
- ✗ No clear objective or benefit.
- ✗ Spelling/grammar mistakes.
- ✗ Unrealistic cost or time estimates.

3. How to Pitch an Idea (Process, Preparation, and Structure)

♦ What is an Idea Pitch?

An **idea pitch** is a **short, focused presentation** used to **convince** an audience about your project or innovation.

Example:

You pitch your idea **“AI Voice Assistant for Visually Impaired People.”**

You explain how it works, benefits, and why it’s unique.

◇ Purpose

- To **communicate your idea clearly and quickly**.
- To **gain approval, support, or investment**.
- To **inspire confidence** in your capability.

◇ Process of Pitching an Idea

Step	Explanation
1. Identify Problem	What issue does your idea solve?
2. Design Solution	How does your idea solve the problem?
3. Know Audience	Adjust tone and content based on audience type.
4. Prepare Slides	Use visuals and keywords (not paragraphs).
5. Practice Delivery	Rehearse several times for fluency.
6. Deliver Confidently	Speak with enthusiasm and clarity.
7. Handle Q&A	Stay calm and answer smartly.

◇ Structure of an Ideal Pitch

Part	Duration	Content
1. Opening	30 sec	Start with a hook — a question or fact.
2. Problem Statement	1 min	Explain the real issue.
3. Solution	1–2 min	Present your innovative idea.
4. Market / Use	1 min	Who will benefit and why?
5. Working / Demo	1 min	Show prototype or explain workflow.
6. Benefits & Uniqueness	30 sec	Why is your idea better?
7. Closing	30 sec	Thank and request support.

◇ Preparation Tips

- ✓ Use **5–7 slides only** (clean & visual).
- ✓ Rehearse timing — finish in 3–5 minutes.
- ✓ Avoid reading from slides.
- ✓ Maintain **eye contact and smile**.

◇ Common Mistakes

- ✗ Speaking too fast or monotone.
- ✗ Overloading slides with text.
- ✗ Ignoring audience engagement.
- ✗ No clear benefit in idea.

4. Elements of Speech Delivery – Passion, Poise & Illustrations

◇ (a) Passion

- **Meaning:** Expressing enthusiasm and energy while speaking.
- It **grabs audience attention** and **shows confidence**.

✓ How to show Passion:

- Use **voice modulation** (high & low tones).
- Add **expressions** while explaining.
- Speak with **interest in your topic**.

Example:

“With this Smart Helmet, we can save thousands of lives every year!”

◇ (b) Poise

- **Meaning:** Calmness and control during presentation.
- Poise shows **professionalism and self-confidence**.

✓ How to maintain Poise:

- Keep a **straight posture**.
- Maintain **eye contact** with audience.
- Don’t panic if you forget a line — pause, smile, and continue.
- Use **natural gestures**.

Example:

When a judge interrupts your presentation, calmly reply:

“That’s an important question, sir. Let me clarify...”

◇ (c) Illustrations

- **Meaning:** Visual or story-based examples used to explain ideas.
- Helps make your **speech memorable and clear**.

✓ Types of Illustrations:

1. **Visuals:** Diagrams, images, flowcharts.
2. **Examples:** Real-world or case-based situations.
3. **Stories/Analogies:** Simple stories that connect the concept.

Example:

“Machine Learning is like how a student learns from mistakes — the algorithm learns from data.”

◇ Tips for Good Speech Delivery

- ✓ Be well-prepared (know your topic).
- ✓ Use **simple language**.
- ✓ Keep good **body language** (eye contact, smile).
- ✓ **Practice** with mirror or friends.
- ✓ End with a **quote or thank you note**.

Summary Table

Topic	Key Points	Example
Thesis / Project Report	Written document for completed work.	Face Recognition System.
Technical Proposal	Written plan before starting work.	Smart Helmet Proposal.
Idea Pitch	Oral presentation to convince others.	Smart Bottle Pitch.
Speech Delivery	Style of presenting ideas effectively.	Seminar Presentation.

1. Difference Between Thesis and Project Report

Point	Thesis	Project Report
Purpose	To present original research work.	To show practical implementation or technical project results.
Focus Area	Research-based and theoretical.	Application-based and practical.
Length	Usually long and detailed (100+ pages).	Shorter (30–60 pages).

Structure	Includes Abstract, Literature Review, Methodology, Results, Discussion, and Conclusion.	Includes Introduction, Objectives, Design/Implementation, Results, and Conclusion.
Example	<i>“A Study on AI-Based Disease Prediction Models.”</i>	<i>“Developing an AI App for Predicting Diabetes using ML.”</i>
Evaluation Criteria	Based on originality, analysis, and contribution to knowledge.	Based on execution, innovation, and working output.

2. Difference Between CV and Technical Proposal

Point	CV (Curriculum Vitae)	Technical Proposal
Purpose	To present personal, educational, and career details.	To propose a technical solution or project idea.
Audience	HR, recruiters, or employers.	Clients, funding agencies, or professors.
Content Type	Personal information, education, experience, and skills.	Problem statement, solution, technical details, budget, timeline.
Tone	Personal and formal.	Professional, technical, and persuasive.
Example	<i>Applying for a Data Scientist job.</i>	<i>Submitting a proposal for a Smart Water Monitoring System.</i>

3. Difference Between Thesis Writing and Technical Proposal Writing

Point	Thesis Writing	Technical Proposal Writing
Objective	To report research findings.	To convince someone to approve or fund a project.
Nature	Academic and research-oriented.	Practical and solution-oriented.
Structure	Introduction → Methodology → Analysis → Results → Conclusion.	Problem → Solution → Budget → Timeline → Benefits.
Length	Long and detailed.	Concise (5–20 pages).
Example	<i>“Optimizing Neural Networks for Image Recognition.”</i>	<i>“Proposal for Implementing Neural Network-Based Face Detection in CCTV Systems.”</i>

4. Difference Between Pitch Presentation and Formal Presentation

Point	Pitch Presentation	Formal Presentation
Purpose	To convince investors or judges about an idea.	To share information, report, or findings.
Duration	Short (3–10 minutes).	Longer (15–30 minutes).
Focus	Business value, innovation, and confidence.	Content depth, clarity, and visuals.
Tone	Persuasive and energetic.	Informative and structured.
Example	<i>Pitching a startup idea for funding.</i>	<i>Presenting your BTech final-year project to faculty.</i>

? 5. Difference Between Speech and Presentation

Point	Speech	Presentation
Medium	Mostly verbal (spoken).	Combination of speech + visual aids (slides, charts, etc.).
Objective	To inspire, motivate, or inform.	To explain, demonstrate, or analyze.
Structure	Opening → Body → Conclusion.	Introduction → Content (with visuals) → Summary/Q&A.
Support Tools	None or minimal (microphone only).	Uses PowerPoint, charts, videos, etc.
Example	<i>Speech on “Importance of Innovation.”</i>	<i>Presentation on “AI in Healthcare.”</i>

? 6. Difference Between Passion, Poise, and Illustration in Speech Delivery

Element	Meaning	Importance	Example
Passion	Energy and enthusiasm in speech.	Keeps audience interested and motivated.	Speaking excitedly about your startup idea.
Poise	Confidence and calm body language.	Builds trust and professionalism.	Maintaining eye contact and posture while speaking.
Illustration	Use of examples, visuals, or stories.	Makes content easy to understand and memorable.	Showing real data chart in a technical talk.

7. Difference Between Written Communication and Oral Presentation

Point	Written Communication	Oral Presentation
Form	Written (text, reports, proposals).	Spoken (speech, presentation).
Permanence	Permanent record.	Temporary, but impactful.
Feedback	Delayed.	Immediate.
Tools Used	MS Word, Google Docs, etc.	PowerPoint, Mic, Projector, etc.
Example	<i>Submitting a project report.</i>	<i>Presenting project results in viva.</i>